Lesson plan · for the teacher

Train, test, trust?

AI · Level 3 · Lesson 3

Time
40 min

Materials
Board and chalk; A card with ten area questions and their answers (for the memoriser); Five new rectangles drawn or written for the test

Learning goals

- Explain overfitting, and why a model's score must come from examples it never trained on.
- Tell false positives from false negatives, and say which costs more in a school case.
- Question an accuracy claim: tested on whom, on which data, with which mistakes?

Activity: Memoriser and learner

1. Warm-up (5 min): Ask: have you met a pupil who memorised last year's paper? How did they do on new questions? Collect two or three stories.
2. Memoriser and learner (12 min): Give one pupil the answer card to memorise for three minutes; teach a second pupil "length times width". Test both aloud on the ten old questions, then on five new rectangles. Keep score on the board.
3. Overfitting (6 min): Name it: the memoriser overfitted. Explain that this is why a model is scored on a test set it never trained on, and that a big gap between the two scores is the warning sign.
4. Two mistakes (9 min): Read out six cards (a mobile-money app blocking a genuine payment as fraud, an eye test missing a pupil who needs glasses, a spam filter letting a scam through…). Pairs sort them into false alarms and misses, then say which costs more in each case.
5. Wrap-up (8 min): Write the three questions on the board: tested on whom? On which data? With which mistakes? Ask the class to apply them to the "98% accurate" claim. Then give the exit ticket.

Prepare a card with ten rectangles and their areas (for example 3 by 4: 12), five new rectangles, and six mistake cards: three false alarms and three misses.

Discussion

- Where do pupils in our school memorise instead of learning?
- In a dropout warning, is a false alarm or a miss worse? Why?
- Which AI claim have you heard that deserves the three questions?

Exit ticket

- Explain overfitting in one sentence, and write one false positive and one false negative from school life.

Try it at school tomorrow

Play "Memoriser and learner": give one pupil the answers to ten rectangle-area questions to memorise (for example "3 by 4: 12"), with no method. Teach a second pupil the method: length times width. Test both on the same ten questions, then on five new rectangles, in front of the class. Ask: who did better on the old questions, who on the new ones, and why? Name it: the memoriser overfitted.

Practice questions · for you

Train, test, trust?

AI · Level 3 · Lesson 3

1 **A dropout tool said a pupil was fine, but she left school in March. Which mistake was that?**

◯ A false negative ◯ Overfitting ◯ A false positive

2 **A dropout tool scored 95% on a test set from city schools, so it will score about 95% with pupils in a rural school.**

◯ True ◯ False

3 **Match each situation to what it shows.**

A cheating detector flags an honest pupil's essay	• • • •	A false negative
An eye test misses a pupil who needs glasses	• • • •	A misleading accuracy score
99% right on training examples, 60% on new ones	• • • •	A false positive
A tool that always says "no risk" still scores 98%	• • • •	Overfitting

4 **A supplier says its attendance-alert AI is "95% accurate". Which questions should you ask? Choose all that apply.**

☐ How many schools have bought it? ☐ How many pupils at risk did it miss?

☐ Was it tested on pupils like ours? ☐ Was the 95% measured on examples it never trained on?

☐ Does it use the newest AI model?

✎ **Think of an AI tool or claim you have met, in an app, an advert or the news. Which of the three questions would you ask about it first, and why?**

Check your answers

Train, test, trust?

AI · Level 3 · Lesson 3

1 **A false negative**

The tool said "no risk" (negative) and was wrong: a missed case is a false negative.

2 **False**

A test set only tells you about pupils like the ones in it.

3 **A cheating detector flags an honest pupil's essay — A false positive**
An eye test misses a pupil who needs glasses — A false negative
99% right on training examples, 60% on new ones — Overfitting
A tool that always says "no risk" still scores 98% — A misleading accuracy score

A false alarm is a false positive and a miss is a false negative.

4 **Was it tested on pupils like ours?; How many pupils at risk did it miss?; Was the 95% measured on examples it never trained on?**

Who it was tested on, which mistakes it makes, and whether the score came from unseen examples tell you if 95% will hold in your school.